

Sea Power

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ABSTRACT

How does sea power differ from land power? We introduce a formal model that explains several characteristics of sea power. Specifically, (i) sea power is more highly concentrated than land power; (ii) it follows cycles; and (iii) is characterized by periodic intensification of arms competition. We account for these features in terms of the high fixed costs and the capital-intensive nature of naval warfare. We find empirical support for our model using a newly digitized dataset of over two dozen navies during the late 19th and early 20th centuries.

Keywords: Sea power; conflict; formal theory

Introduction

Sea power is the ability of a state to use naval force to influence other states. Since the writings of Alfred Mahan (1890), sea power has been recognized as a crucial factor in great power competition and international relations, and the sea has been regarded as “a strategic highroad, a medium by which one group of people can dominate the affairs of another” (Till, 2013, pp. 14–15). Sea power is also important for global commerce — naval power can help open new

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trade routes, as well as improve the terms of trade with existing trade partners (Glaser and Rahman, 2016; Rahman, 2010). The importance of sea power has long been recognized by naval strategists and military historians (Falk, 2000; Mahan, 1890; Till, 2013). But with the exception of the seminal work of Modelski and Thompson (1988), it has rarely been the focus of sustained analysis by economists or political scientists.

The development of sea power played a crucial role in the rise and fall of numerous world powers including Athens, Venice, the Dutch Republic, Great Britain and the United States. The difference between sea power and land power is also central to many open questions in historical political economy. For example, intensified military competition played a critical role in the formation of European states after 1500 (see Johnson and Koyama, 2017; Parker, 1976, 1988). But this literature has focused on armies and has largely neglected naval power.¹ The relationship between the military and political institutions is also an important topic. However, much less is known about the differences between naval forces and regular armies. These outstanding questions point to the importance of studying sea power through the lens of historical political economy.

In this paper, we investigate sea power theoretically and empirically. First, we establish that sea power differs from land power in the following respects: (i) it is highly concentrated and typically characterized by the existence of a clear leading power; (ii) it is cyclical with long periods of stasis interspersed with episodes of intense competition. Second, we develop a formal model that can account for these stylized factors. This model delivers a number of predictions concerning the incentives that the leading power and its challengers have to invest in sea power. Third, we utilize newly digitized naval data to investigate these predictions during the age of intensified naval competition between 1880 and 1914.

Our model has the following features: (a) states build naval power to control economic resources; (b) success in pairwise competition between states is decided by a contest success function; and (c) the cost of investment in naval power is determined by a state's underlying military capacity. We focus on the case where naval competition is characterized by non-simultaneous investments in military power. In this case, we establish the following results. First, the leader (i.e., the country that has an initial advantage in underlying military capacity) and the follower (i.e., the challenger) increase their investments in naval power in response to an increase in one's own capacity or in the value of the economic resources being fought over. Second, if the military capacity of the leader increases, then the follower decreases its investment in naval power.

¹Glete (2000), for instance, notes "[t]he role of navies in the state formation process has often been neglected".

However, if the capacity of the follower increases, the leader may increase its investment to deter competition, especially if its capacity advantage is large enough (as in this case warding off competitors is relatively inexpensive to the leader). Third, we show that an intensification of naval investments and competition can occur when there is a symmetric increase in military capacity. Finally, we show that in the presence of fixed costs, the leader may “over-invest” in naval power in order to make it unattractive for potential competitors to engage in naval competition.

To test the model, we turn to a unique ship-level dataset from the late 19th to early 20th centuries. Our dataset is constructed from Brassey’s *Naval Annuals*, which are yearly accounts for all measurable naval activities around the globe. This period saw a number of important innovations that contributed to making ships larger, more expensive, and more powerful, including the development of armored battleships, explosive shells, ironclads, the Dreadnought, and naval aviation. We show that all navies in this period increased investments in newer and larger ships as their military capacities grew, and as the value of sea-based trade increased. We also discuss evidence on how spending by the Royal Navy in general deterred spending by other navies.

Relationship with the Literature The purpose of sea power is to influence political and military affairs on land (Till, 2013). The concept of sea power as a strategic arm and as a means of controlling international resources owes its origins to Alfred Mahan (1890). Following Mahan, the importance of sea power has been widely recognized by naval strategists and military historians (Corbett, 1911; Falk, 2000; Lambert, 2018; Till, 2013). Theorists of geopolitics have contrasted the significance of sea power with that of land power (see, especially, Mackinder, 1904).

In the wider historical literature, the rise and fall of British sea power in particular has attracted numerous scholars and popular treatments (see, in particular Kennedy, 1976). Particular episodes of naval history have also received numerous in-depth studies, especially the Anglo-German naval race from 1898 to 1914 (Iambi, 1984; Padfield, 1974; Rose, 2007; Seligmann, 2015, 2016) and the British navy in the Age of Sail (e.g., Dull, 2009; Marcus, 1971).² But there has been relatively little research by social scientists on the rise and fall of naval powers.

Recent work by political scientists, economic historians, and economists typically uses naval data to test more general propositions and theories. For example, Lindsey (2019) uses data on British naval battles between 1650 and

²There are a large number of historical surveys that focus on military aspects of sea power (see Gray, 1992; Lambert, 2018; Lewis, 1951; Padfield, 1982; Richmond, 1934; Rose, 2007). There have also been studies of the role of sea power in ancient history (Shepard, 1925; Starr, 1989) and of the rise of American sea power (see Sprout and Sprout, 1939).

1833 to test the theory that mutual optimism causes international conflicts. Specifically, he shows that when naval battles were the result of an inadvertent coming together of two navies, observable measures of capability do not predict victory. However, when the battle was forced by one side, these indicators do predict the victor. Allen (2002) and Benjamin and Thornberg (2007) examine the institutions that made the British navy successful. Voth and Xu (2020) study patronage institutions within the British navy. They show that the patronage system was surprisingly successful in selecting and rewarding naval commanders.

Historians have written numerous assessments on the relative strengths and strategic positions of the world's major sea powers. But there is relatively little research on naval power in economic history or political science. Quantitative accounts that trace the rise and fall of the major sea powers are also rare. The most systematic study is Modelski and Thompson (1988) who provide estimates for the size and strength of the world's major navies since the Age of Discovery. This comparative neglect of sea power in economic history and political science is surprising, since naval power played a crucial role in the rise of Western economies from 1500 onwards, in the Great Divergence, and in the spread of globalization (Findlay and O'Rourke, 2007).³ The Dutch Republic, Great Britain, and the United States were all, at different times, both economic powerhouses and naval leaders.⁴

Our empirical analysis focuses on this period of rapid naval investment and geopolitical competition. In so doing we build on a small number of papers that have investigated the role sea power played in the first era of globalization (1870–1914). Rahman (2010) uses a gravity model to explore how either allying or fighting with a naval power helps or hinders a country's interstate commerce. Glaser and Rahman (2016) introduce new archival data to demonstrate that sea power had a causal effect on the direction of trade. Both Rahman (2010) and Glaser and Rahman (2016) establish that sea power had important economic implications in the pre-World War I period, but they do not seek to explain differences between or changes among naval powers, which is the primary focus of this paper.

³Findlay and O'Rourke (2007) do emphasize the importance of military power in shaping trade routes and economic activities. They also document the role that sea power played in first Dutch economic leadership in the 17th century (pp. 238–239) and then Britain's subsequent rise to prominence (pp. 240–284).

⁴As one historian puts it: “[t]he history of the West, hence of the modern world, has been written by the sea powers; it was their market-acquisitive system that encircled and dominated the globe, opening up new lands for new cash crops and commodities, sinking mines, building foundries and mills and stimulating technological change” (Padfield, 1982, p. 3).

Stylized Facts about Sea Power

To motivate our analysis we first consider several stylized facts that stand out from the history of sea power.

Sea power is unequally distributed This is evident both today and historically. Figure 1 depicts the share of the largest country's land or sea force as a share of the total land and sea forces.⁵ Today, the largest land power (China) makes up only around 11% of the world's total land armies. In contrast, the world's largest sea power (USA) makes up almost 35% of the world's total naval power. We also report in Figure 1 the sum of squares of each individual land and navy power's share. This is a measure of concentration that is equivalent to the Herfindahl–Hirschman Index (HHI) used to measure industry concentration. If one power controlled all of the land or sea forces available, the HHI would be 1. We calculate that the HHI for naval power is 0.164, or more than four times higher than land power (0.037).

Indeed this skewed distribution of sea power is common historically. One implication of this stylized fact that will motivate our theoretical model is that there is typically a single dominant naval leader.

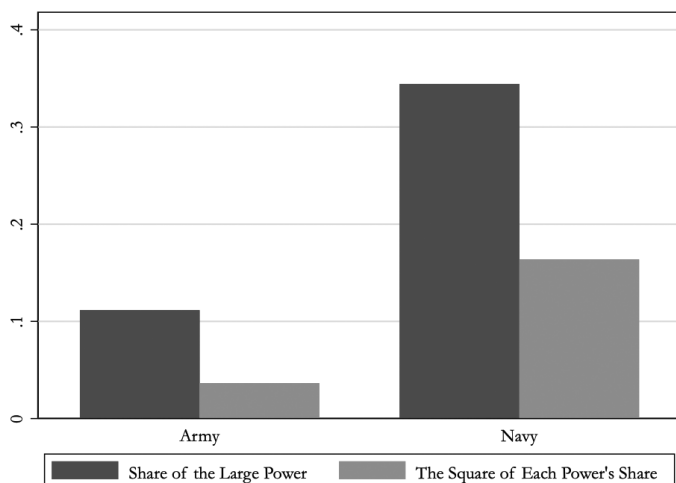


Figure 1: The relative concentration of land and sea based forces today.

This figure depicts the share and the shared squared of the largest army in the world (China's) and the largest navy (the USA's) as a share of the total land and sea forces.

Source: Wikipedia.

⁵This data is available from Wikipedia. See <https://tinyurl.com/dxacx57> and <https://tinyurl.com/qx7nju8>.

Comprehensive data on historical armies is not available and not directly comparable. In particular, the size of armies changed dramatically between peacetime and wartime.⁶ In contrast, as a navy took much longer to build, the size of a fleet tended not to vary dramatically between peace and war times. Nonetheless, we are able to establish that this stylized fact also holds historically using two different approaches. First, we can obtain estimates of army size for selected countries and selected years — specifically, on the eve of major wars in 1740, 1800, and 1914 — and use this data to estimate the HHIs for land powers for these years (see Table 1). For 1914, we have data on 18 major powers and the HHI is 0.11. The largest army prior to World War I belonged to Russia, but it represented less than 20% of total land forces. The corresponding HHI for naval powers is 0.27, with the United Kingdom alone accounting for 44% of the world’s battleships. The HHI for land powers is 0.09 for 1800 and 0.13 for 1740, and in both cases much lower than the corresponding HHIs for naval strength.

The second approach builds on the fact that the maximum land army that could be mobilized was usually a function of the total population.⁷ We can therefore estimate HHIs for European populations for the entire period between 1600 and 1914. This data is depicted in Figure 2. It provides a good estimate of the potential land army that could be mobilized. Both measures indicate that the naval power has always been much more concentrated than land power.

Sea power follows cycles. While naval strength has always been highly concentrated compared with land power, the degree of concentration has varied

Table 1: Army and navy concentrations over time.

	Time Period		
	1740	1800	1914
Army HHI	0.130	0.094	0.11
Naval HHI	0.295	0.265	0.27
Army HHI as a % of Naval HHI	44%	35%	41%

Authors’ own calculations. Data Sources: See Appendix B.

⁶For example, the English army in 1697, at the conclusion of the Nine Years’ War, was 100,000 strong. But it was reduced to just 20,000 by 1700, just a few years after peace prevailed.

⁷The largest army in the world until the 20th century belonged to Qing China — the most populous country in the world. The largest army in Europe from the 17th to the early 19th century was the French which was also Europe’s most populous nation for much of that period. There are notable exceptions to this generalization such as the Roman Republic, Sweden in the 17th century, and Prussia in the 19th century.

over time. There have been episodes in which the dominant naval power has more or less been unchallenged — Britain during the mid-19th century and the United States after 1989 — and periods of convergence and intense attempts by followers to catch up with the leader.

In the 16th century, Portugal and then Spain were the dominant naval powers.⁸ But Portuguese naval power was challenged by the Netherlands in the first half of the 17th century. Dutch naval power was then challenged by England and France at the end of the 17th century.⁹ The British navy rose to supremacy in the 18th century, though at several points thereafter the French (and Spanish) navies threatened to overturn Britain's dominance at sea. Britain maintained its naval supremacy in the middle decades of the 19th century, largely because it was not directly challenged by other powers. In the 20th century, naval dominance passed from Britain to the United States during World War II.¹⁰

It is evident from the history of naval power that achieving and maintaining naval supremacy requires continuous investment. Spanish naval power collapsed in the late-17th century.¹¹ The efforts required to maintain British navy supremacy in the late 18th century were tremendous. O'Hara (2010, p. 116) notes that "this huge naval endeavor cost a great deal, for it was a highly complex and technological effort. The amount of money that went into building a navy ship was much higher than the cash stocks invested in the new industries of the 'industrial' revolution". By the end of the Napoleonic Wars, Britain had successfully defeated all threats to its naval supremacy. Naval historians nonetheless describe the British navy as suffering from underinvestment and a lack of new ships in the decades following 1815.

In contrast, the size of the largest land army was often stable over time. As noted above, this was perhaps because before the 20th century, army size was largely a function of the total population size. France with around 20 million inhabitants in 1700 was the most populous country in Europe and the French army was by far the largest in Europe. It was only in the 19th century that the population of Russia exceeded France's significantly, and it was in the 19th century — following the defeat of Napoleon in 1815 — that the Russian army overtook the French as the largest in Europe, at 900,000 strong by 1850.

⁸Between 1580 and 1640, Portugal and Spain were ruled by the same monarch.

⁹This history is extensively documented by Kennedy (1976) and Herman (2004). Note that we refer to England prior to the Act of Union of 1707 and Great Britain thereafter.

¹⁰Already in 1881, however, the United States had committed itself to becoming a major naval power (Sprout and Sprout, 1939, p. 183). By 1897, the US navy had a fleet which though smaller than that of its European rivals could "hold indisputable command of all the strategic marine approaches to the continental United States" (Sprout and Sprout, 1939, p. 222).

¹¹Mahan (1890) dated the decline of Spanish naval power to the Battle of the Downs in 1639 and the subsequent historical scholarship largely seconded the view that the Spanish navy struggled to rebuild its strength following this defeat (Strading, 1979, p. 208).

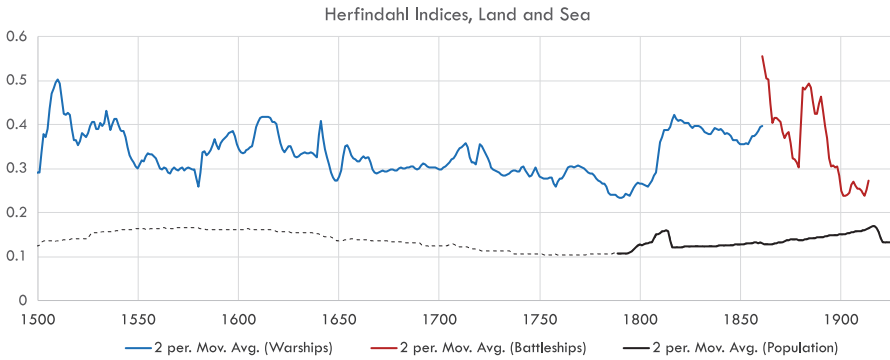


Figure 2: The concentration of naval power and land power over time. Information on naval power concentration is calculated from data in Modelski and Thompson (1988). Population information is from McEvedy and Jones (1978) and Maddison (2003).

The Russian army remained the largest army in Europe until World War I.¹² This is evident in Figure 2, which shows that the distribution of populations across European countries is more stable over time than the distribution of naval power.

A Model of Sea power Competition

In this section, we introduce a simple model to study sea power. The model builds on the literature that uses contest success functions to study conflict (Garfinkel and Skaperdas, 2007; Hirshleifer, 2001; Skaperdas, 1996). In order to focus on cases that are relevant for understanding the dynamics of sea power historically, we employ a Stackelberg game. There is a dominant power, or Stackelberg leader, and a secondary power who acts as the Stackelberg follower.

Setup

Consider two countries, 1 (leader) and 2 (follower), engaged in a two-stage Stackelberg game. The timing of the game is as follows.

1. Country 1 chooses its naval investment x_1 .
2. Upon observing x_1 , Country 2 then chooses its naval investment x_2 .

¹²Estimates of historical populations are provided by McEvedy and Jones (1978).

3. The choice of x_1 and x_2 jointly determine how a prize V is shared between them.

The payoffs of the two countries are given by

$$\pi_1(x_1) = \frac{x_1}{x_1 + x_2} V - \theta_1 x_1^2 - \phi \quad \text{and} \quad \pi_2(x_2) = \frac{x_2}{x_1 + x_2} V - \theta_2 x_2^2 - \phi, \quad (1)$$

respectively, where θ_i inversely measures the underlying military (naval) capacity of country i , or the degree of ease for country i to increase its military (naval) investment, ϕ (≥ 0) is a constant that denotes the fixed cost of naval investment.

This model captures the difference between land- and sea-based competitions in three ways. First, our choice of Stackelberg rather than Cournot competition reflects the fact that naval competition is typically characterized by a leader (Stylized Fact 1). Given that the leader sets the benchmark for other competitors, we model it as the first mover.

Of course, land warfare could also have a sequential structure, i.e., in land warfare there could also be a dominant power which has invested in its military capacity in the past. Our contention, however, is that this sequential structure is *particularly* true of naval competition because naval competition is more capital intensive. In particular, naval investment are more irreversible in the sense of Dixit and Pindyck (1994) because they cannot as easily be repurposed.

Second, to capture the leader's advantages in building naval power, we assume that $\theta_1 < \theta_2$. Here θ *inversely* measures the underlying military capacity, broadly conceived. The assumption implies that the leader could build naval power at a lower cost than its rivals. This assumption is based on the observation that investment in naval power typically required both access to the coast and a merchant marine to provide knowledge of seamanship and trained crews. For example, the Dutch advantage in shipbuilding in the early 17th century, described by Harding (1999, p. 15), corresponds directly to what we mean by a low θ . He refers to the "[t]he concentration of capital, mercantile experience, expertise in exploiting wind power, labour-saving devices and manipulating wood were easily transferred to the shipyards" in explaining why "Dutch builders were able to produce vessels 40 to 50 percent cheaper than it was possible in England."

When comparing sea-based competition with land-based competition, it is natural to assume that the difference in military capacity between countries is greater for sea-based competition than for land-based competition: $|\theta_1^{Sea} - \theta_2^{Sea}| > |\theta_1^{Land} - \theta_2^{Land}|$. In practice, this amounts to granting that, for instance, countries with extensive coastlines like Great Britain are more able to scale up their investment in naval power than are more landlocked countries like the Habsburg Empire.

Third, we introduce a fixed cost ϕ to capture the fact that naval investment is capital intensive. Specifically, naval warfare is much more capital-intensive than land-based warfare. The capital intensity of land-based conflict varies over time. Following the invention of gunpowder and the rise of artillery, warfare in general became more capital-intensive. But for land-based conflicts it is always possible to simply augment one's army with more soldiers, whereas this is *not* the case with naval conflicts as ships are essential in such warfare.¹³

While the fixed costs of naval investment varied over time, constructing a fleet was always expensive.¹⁴ While armies could be expanded rapidly in times of war, navies capable of offensive actions required years of investment and planning. Specifically, Harding (1999) reports that in the mid-18th century the cost of building a 60-gun man of war, "the smallest vessel that could, by then, stand in the line of battle," was £23,531. The cost of a larger ship such as the 100-gun Victory was £63,174 and this cost "excludes all costs related to victualling, storing, ordnance and wages required to get the ship to sea." Refitting the Victory cost between 50% and 120% of the original cost of the ship. In contrast, he notes that "[t]he entire maintenance cost of a regiment of foot for one year in 1762 amounted to £18,365."

Dutch sea power in the 17th century was funded by the highest level of taxation in Europe. Similarly, paying for the Royal Navy dominated the budgets of English governments throughout the 18th century (Brewer, 1988). The introduction of steamships and later Dreadnoughts in the late 19th and early 20th centuries raised the fixed costs to establish a navy even higher.

Comparative Statics

We derive five propositions by solving the game by backward induction. For ease of exposition, we assume that the fixed cost ϕ is zero in the first four propositions and consider the case of a positive ϕ only in the last proposition. Proofs are provided in the Appendix.

Proposition 1 (Capacity and Investment). *The Stackelberg leader and the follower increase their own naval investment in response to an increase in its naval capacity, i.e., $\frac{dx_1}{d\theta_1} < 0$ and $\frac{dx_2}{d\theta_2} < 0$.*

Proposition 2 (Prize Enhancement and Naval Competition). *An increase in the size of the prize induces both players to increase their naval investments, i.e., $\frac{dx_1}{dV} > 0$ and $\frac{dx_2}{dV} > 0$.*

¹³In 18th-century Britain the cost per active fighting man in the navy was twice that of the army (O'Hara, 2010, p. 116).

¹⁴Thus "[i]n times of peace the cost of maintaining a naval force was so high that it was politically difficult to sustain" (Harding, 1999, p. 38).

Proposition 3 (Asymmetric Growth).

- a *The Stackelberg follower decreases its investment in response to an increase in the leader's naval capacity, i.e., $\frac{dx_2}{d\theta_1} < 0$.*
- b *The Stackelberg leader increases its investment in response to an increase in the follower's capacity iff the gap in capacity between the leader and the follower is sufficiently large, specifically, $\frac{dx_1}{d\theta_2} < 0 \iff \theta_1 < \frac{2}{3\sqrt{3}}\theta_2$.*

This proposition suggests that if the Stackelberg leader enjoys substantial advantage over the follower in its ability to scale up investment, in response to a strengthening of the follower's underlying military capacity, the leader will in fact increase its investment in the first stage, so as to deter and discourage the follower from making a large investment in the second stage.

Proposition 4 (Symmetric Growth and Naval Competition). *If the naval capacities increase from θ_1 to θ'_1 and θ_2 to θ'_2 such that $\frac{\theta_1}{\theta_2} = \frac{\theta'_1}{\theta'_2}$, then naval investments increase from x_1 to x'_1 and x_2 to x'_2 such that $\frac{x_1}{x_2} = \frac{x'_1}{x'_2}$.*

This proposition states that a symmetric growth in the underlying capacities of the leader and the follower induces a symmetric growth in their naval investments. For instance, a technological advancement shared by all countries, or an across-the-board decrease in the cost of raising tax (e.g., economic growth in all competing countries), will lead to a general increase in naval spending, thereby intensifying military competition.

Proposition 5 (Fixed Cost and Over investment). *There exists $\hat{\phi}$ such that, if $0 \leq \phi \leq \hat{\phi}$, the Stackelberg leader's level of investment is weakly increasing in the fixed cost, i.e., $\frac{dx_1}{d\phi} \geq 0$. Now consider ϕ° and ϕ' such that $0 \leq \phi^\circ < \phi' \leq \hat{\phi}$, let x_1°, x_2° and x'_1, x'_2 denote the optimal levels of naval investments when $\phi = \phi^\circ$ and $\phi = \phi'$ respectively, if $x'_1 > x_1^\circ$, then $x'_2 = 0$.*

The proposition states that when the fixed cost of participating in naval competition is non-zero, the Stackelberg leader may have the incentive to increase its naval investment to an “excessively” high level, so as to deter others from entering into naval competition with the leader. Furthermore, the higher the fixed cost ϕ , the greater the incentive for the Stackelberg leader to overinvest. This is because a high fixed cost makes it costly for the follower to enter into competition. By increasing its investment marginally, the leader may be able to convince a competitor that already finds the high fixed cost a significant dampener to give up competing entirely.

The proposition provides a potential explanation as to why leading naval powers, historically and today, sometimes focus on building a naval force that seems excessive in light of the investments made by their potential competitors.

Our model suggests that acts of overinvestment may be rationally calculated to deter competition. While it may appear irrational for the leader to spend heavily on the navy when its competitors appear unenthusiastic in engaging in naval competition, it is precisely the high spending of the leader that discourages its competitors from spending more.

A Numerical Example

For ease of illustration, we have analyzed a two-player game. This model can be extended to include more players. We now provide a numerical example that extends the model to three players. We have in mind (1) a dominant power, (2) a rising power that plays catch-up, and (3) a peripheral player that stays weak throughout. This numerical example illustrates how the decisions of the naval leader impact the naval investment decisions of its rivals.

The timing of the game is as follows. In stage one, Country 1 chooses its naval investment x_1 . In stage 2, upon observing x_1 , Countries 2 and 3 simultaneously and independently choose their naval investments x_2 and x_3 . The decisions of x_1 , x_2 , and x_3 jointly determine how the prize V is shared between the three countries.

Let the prize $V = 1$, and the fixed cost of naval investment $\phi = 0.05$, the naval capacities $\theta_1 = 1$ and $\theta_3 = 20$. Initially, $\theta_2 = 20$ too. We vary it from 20 (parity with Country 3) to 1 (parity with Country 1) to see how it affects the distribution of naval strengths (x_1^* , x_2^* , and x_3^*) and shares of spoils (V_1^* , V_2^* , and V_3^*).

Figure 3 illustrates the results of this exercise. Figure 3(a) shows that when there is a large gap between the naval capacities of the leader and the followers, the leader will invest significantly in the navy. On the surface, the leader's investment appears excessive since the followers are investing nothing, but it is precisely the high investment of the leader that keeps its followers at bay.

When the capacity of Country 2 strengthens initially (θ_2 decreases from 20 toward 14), Country 1 makes an increasingly large investment to discourage Country 2 from entering the naval contest. But once θ_2 falls below around 14, Country 1 reduces its investment in naval power because it is no longer beneficial for Country 1 to keep Country 2 completely out of the naval contest. From then on, the HHI drops sharply to below 1 as Country 2 makes a strictly positive investment to challenge Country 1's supremacy.

Figure 3(b) shows the distribution of the prize V among the three countries. Of particular interest is the deadweight loss from naval competition. The deadweight loss is incurred because the three countries could theoretically negotiate a deal in which each country will make zero naval investment and the prize V is shared among them such that no country is worse off and some or all countries are better off than if they competed strategically. But in practice

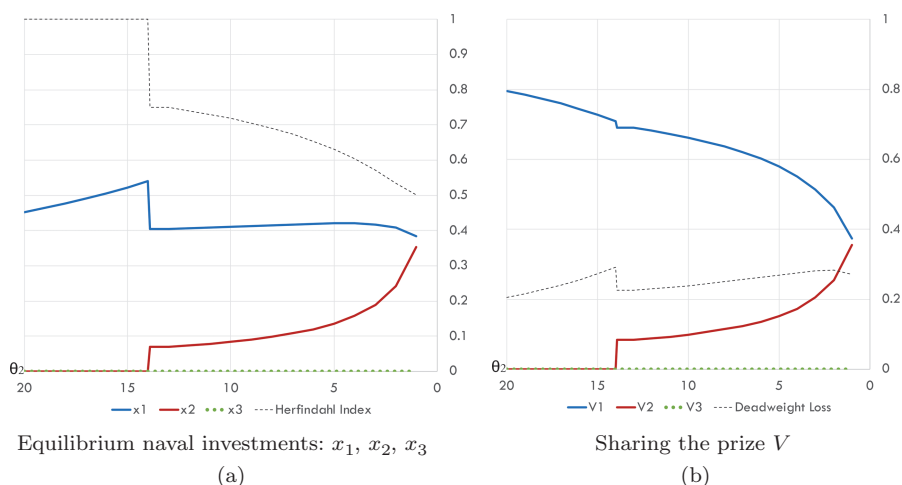


Figure 3: A numerical example.

We consider a contest of three players: (1) a dominant power ($\theta_1 = 1$), (2) a rising power that plays catch-up, and (3) a peripheral player that stays weak throughout ($\theta_3 = 20$). Figure (a) depicts the equilibrium naval investments x_1 , x_2 , and x_3 as θ_2 falls from 20 to 1. Figure (b) depicts how the prize V is shared among the three countries.

such a deal is impossible because the countries cannot credibly commit to it. Note that the deadweight loss that arises from naval investment is highest when Country 2 is at the doorstep of entering the contest (i.e., around and above $\theta_2 = 0$) and when Country 2's underlying military capacity approaches that of Country 1.

Applying Our Model to the First Age of Globalization

To test our model's main predictions more systematically, we focus on the late 19th century, the First Age of Globalization, which was a period of growing international trade and naval competition. In the late 19th century, Britain was the global naval hegemon. But it was a period of technological innovation and expanding trade, when rival powers such as the United States, France, and Germany began to challenge Britain's position.

Together the predictions of our model shed light on many features of naval competition. Proposition 1 states that if a country is engaged in naval competition, then it always responds to an increase in its underlying capacity by increasing its military investment so as to capture a larger size of the prize. This formalizes the idea that an increase in underlying military capacity, such as what the Dutch experienced in the late 16th and early 17th centuries or the English after 1650, will led to greater naval investment. Similarly the relative

decline of the Spanish navy after 1650 was precipitated by the decline in their merchant and commercial fleets.

Proposition 2 states that a rise in the value of sea-borne trade should induce more investment in naval power. The broad historical evidence is consistent with this. The rise of Atlantic trade after 1500 saw European powers invest in much larger ships than previously. Similarly, trade expansion during the First Era of Globalization was associated with what historians term “the age of navalism” (Rose, 2007).

To test Propositions 1 and 2, we need empirical proxies for (i) naval strength x ; (ii) the value of sea-based trade V ; and (iii) naval capacity θ .

To obtain a rich set of measures of x , we digitize naval power and expenditure measures for two dozen countries documented in Brassey’s Navy Annuals. Data include each naval vessel’s name, class, displacement, horsepower, armament, ordnance capability, speed, and other characteristics. We construct a panel dataset that provides these measures across navies for the period 1886–1913. Details of data construction are available in the Appendix.

Figure 4 uses this data to illustrate the investment in sea power made by the leading naval powers (Great Britain, United States, Russia, France, Japan, and Germany) in the late 19th and early 20th centuries.

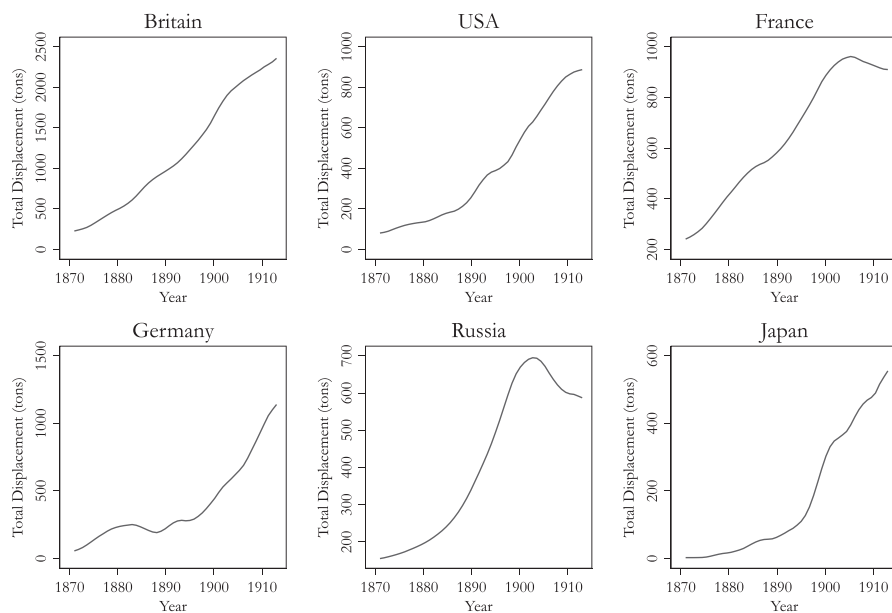


Figure 4: The rise of sea power in the First Age of Globalization.

This figure plots total displacement for the six major sea powers in the early 20th century. For details on data construction, see main text.

Table 2: Summary statistics of 6 naval powers, 1886–1916.

Naval power	Variable	Mean	St.dev.	Min	Max
Britain	No. of ships	425.3	63.1	315	542
	Total displacement (thous. tons)	1694.4	599.2	795.5	2647.4
France	No. of ships	153.8	34.9	57	195
	Total displacement (thous. tons)	809.0	169.3	543.0	1054.3
Germany	No. of ships	99.3	22.9	63	134
	Total displacement (thous. tons)	582.4	368.6	180.8	1362.88
United States	No. of ships	100.9	16.7	67	132
	Total displacement (thous. tons)	577.4	260.9	186.1	965.2
Russia	No. of ships	101.6	23.2	37	140
	Total displacement (thous. tons)	548.1	162.5	257.9	806.2
Japan	No. of ships	52.8	16.0	28	73
	Total displacement (thous. tons)	303.7	206.1	54.9	656.1

Table 2 presents the summary statistics for the top naval powers. Together with Figure 4, it makes clear that Britain was the naval leader. We also see tremendous growth in overall naval power, with particularly explosive growth for Germany, Japan, and the United States. As battleships made way for pre-Dreadnoughts, which in turn made way for Dreadnoughts, ships grew dramatically in size and power.

To obtain an empirical measure of V , it is important to recognize that the value of trade may be endogenous. For example, in studies assessing the impacts of trade on economic prosperity (Helpman, 1988; Rodrik, 1995; Romer and Frankel, 1999), inference is complicated by the fact that countries whose incomes are high for reasons other than trade may trade more. In our case, countries which are able to build and maintain naval fleets are better equipped to establish extensive trade routes and secure better terms of trade for their products (Glaser and Rahman, 2016; Rahman, 2010). Any estimation procedure regressing naval power on trade may therefore suffer from reverse causality or omitted variable bias.

In order to address these concerns, we construct an estimate of sea-based trade. To do this we re-estimate the bilateral trade model in López-Córdova and Meissner (2008), but distinguish between sea-borne and land-based trade. This requires estimating a bilateral trade model between country-pairs from

1870 and 1910, controlling for economic size, distance, and other country level features typical in gravity models, but exclude all bilateral pairs whose trade was conducted primarily by land routes. Another important difference here is that we focus on the economic size and strength of the trading *partner*, thereby capturing factors that potentially affect trade that are unrelated to the economy of our country of interest (see Romer and Frankel, 1999). We aggregate each predicted value of bilateral trade volume for each country each year.¹⁵

Of course one could imagine cases where the change in a state's GDP is caused by a trading partner's naval capacity. For example, a local ally might gain economically from national defense spillovers originating from the naval power. One could also imagine the opposite occurring for potential rivals. In any case, by aggregating across all trading partners, we limit the potential of naval power spending to impact the overall income of trading nations.¹⁶ By using estimated trade flows instead of actual trade flows, we accomplish two things — we limit measurement error by isolating only seaborne trade, and we mitigate the potential endogeneity of naval power itself influencing trade.

Table 3 reports the results of this bilateral gravity model. As expected, a trading partner's GDP and population positively predict a country's trade volume with that partner, while distance from the trading partner negatively predicts trade volume. These factors, important for commerce between a country and its trading partner, are arguably exogenous to that country's decision to invest in naval power. Column (1) reports a parsimonious specification that includes only GDP, population, and distance. Our preferred specification includes cultural and geographical factors that also predict trading relationships. In our main analysis, we employ the estimates reported in Column (2) of Table 3 as our preferred proxy for sea-based trade. In Column (3), we exclude trade with Great Britain, as Britain was the leading trading nation during this period. Reassuringly, the size of most coefficients remains largely unchanged.

In order to capture θ — which measures underlying naval capacity in our model — we employ either GDP, or a national capacity index for industrial production provided by the Correlates of War database. We also control for the total size of the navy as measured by total displacement of all commissioned vessels, and include a time trend in all specifications.

The resulting dataset is an unbalanced panel that spans 1870–1914 and includes 29 countries. As it is a long panel, we check for unit roots in both the dependent and explanatory variables. We first difference the non-stationary

¹⁵Data from López-Córdova and Meissner (2008) are measured in 5-year increments. We take our 5-year trade estimates and linearly interpolate for in-between years.

¹⁶Alternative first-stage specifications where we exclude the GDP of trading partners yield similar second-stage results.

Table 3: Determinants of sea-based trade.

Dep. Var:	Log sea-based trade _{ij}		
	(1)	(2)	(3)
Log Real GDP _j	1.448*** (0.0532)	0.587*** (0.0231)	0.501*** (0.0238)
Log Population _j	0.457*** (0.0246)	0.416*** (0.0102)	0.385*** (0.0106)
Log Distance _j	1.983*** (0.0319)	−0.517*** (0.0161)	−0.245*** (0.0174)
Island _{ij}		0.525*** (0.0287)	−0.914*** (0.0369)
Common Language _{ij}		0.529*** (0.0494)	0.265*** (0.0507)
Landlocked		−1.020*** (0.0458)	−0.751*** (0.0450)
Excluding trade with Great Britain			✓
Observations	17,518	17,518	15,187
pseudo R ²	0.197	0.862	0.878

variables in order to conduct our empirical analysis. Further details and the results of the relevant panel unit root tests are available upon request.

Our empirical strategy involves regressing new naval spending (a flow variable) on the change of the existing stock of the country’s naval capital, along with other factors. We estimate:

Naval Investment_{i,t} = α + β₁ΔNaval Capacity_{i,t} + γX' + Λ_i + Γ_t + ε_{i,t},

(2)

where Γ_t are year fixed effects and Λ_i are country fixed effects. X includes a measure of underlying naval capacity and our aggregate measure of predicted trade, ∑_j Sea-based Trade (using the estimates described above).

The approach we take is to test whether the patterns in the data that we observe are *consistent* with the predictions of our theoretical framework. Given data limitations, we cannot fully rule out all alternative explanations. But the results establish that our model does provide a coherent account of naval investments during the “age of navalism.”

The results that we report in Table 4 indicate that the capacity to build naval capital, be it captured by GDP (Column 1) or Correlates of War’s

Table 4: Determinants of naval spending: Testing Propositions 1–2.

	(1)	(2)	(3)
Dep. Var:	New total displacement		
GDP	0.786*** (0.0389)		
Nat. Capacity Index		6.774*** (0.424)	4.517*** (0.373)
$\sum_j \hat{\text{Sea-based Trade}}$			0.0107*** (0.000706)
Δ Displacement	0.000927*** (0.0000676)	0.00100*** (0.0000723)	0.000843*** (0.0000613)
Observations	1,256	1,144	1,025
Pseudo R^2	0.113	0.098	0.123

Note: * $p < .10$, ** $p < .05$, *** $p < .01$.

National Capacity Index (Column 2), is positively associated with naval capital investment.¹⁷ Sea-based trade also predicts naval spending (Column 3).

Now we turn to Propositions 3–5. Propositions 3–4 shed light on the distribution of sea power. The first part of Proposition 3 states that if the leader strengthens in naval capacity, the follower will reduce its investment. The second part of Proposition 3 states that so long as the capacity gap between leader and follower is sufficiently large, a strengthening of the follower's capacity will lead to the leader investing more. Together with Proposition 4, these results provide guidance on the conditions under which we would expect an intensification of naval competition.

Note that Propositions 3–5 do not lend themselves to a direct empirical test as we lack a source of exogenous variation in naval spending. Nonetheless, we can employ our data for the period 1880–1914 to illustrate these propositions.

First, Proposition 3(a) helps to explain why until the late 19th century few countries sought to challenge the naval leadership of Great Britain. In 1870, total British naval displacement was 122,006 tonnes. It steadily increased through the 1870s and 1880s, reaching 491,578 tonnes by 1880. This was driven by both the size and number of ships. By 1890, it had almost doubled again to 971,359 tonnes. There was another dramatic increase during the first decade of the 20th century. In 1905, total British displacement was 1,956,300 tonnes. Following the launch of the first Dreadnoughts, it rose again reaching 2,607,539 by 1914. This was driven by both an increase in the number of ships

¹⁷The correlation between National Capacity Index and GDP is approximately 0.82.

from 117 in 1878 to 455 in 1900 and 512 by 1914 and in their average size: from around 3,905 tonnes in 1878 to 5,093 tonnes by 1914.

In contrast, the navies of Britain's rivals remained small during the 1870s and 1880s. Total German displacement in 1880 was 239,549 tonnes or less than 50% of British displacement. It did not increase in the 1880s. As a consequence, by 1890 it had fallen to only 198,729 or 20% of British displacement. Major German naval investment only began with the onset of the Anglo-German naval race (see Figures 5(a) and 5(b)).

There were no major naval challenges to Britain until the 1890s.¹⁸ But in this decade, new naval technologies began to become available including armored battleships, explosive shells, ironclads, the Dreadnought, and, later, naval aviation. The construction of the HMS Dreadnought in 1906, for example, dissuaded many lesser powers from attempting a similar expansion as predicted by Proposition 4. This is evident in the naval investments made by several major powers including Germany, the United States, and Japan (see Figure 4). In particular, it prompted German Admiral Alfred von Tirpitz to accelerate an already higher rate of German naval construction (Marder, 1961). The German Fleet Acts passed between 1898 and 1912 saw substantial increases in naval investment. Focusing on Germany as a paradigmatic follower, we can see that in Figure 5(b), the ratio of German displacement to British displacement increased from 1898 onwards and especially accelerate after 1906.

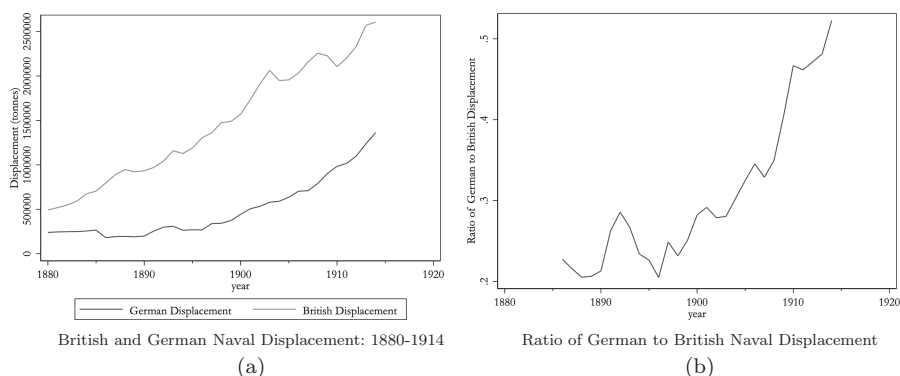


Figure 5: Anglo-German naval competition.

Source: See main text.

¹⁸According to Kennedy (1976), the first signs that Britain's supremacy was to wane appeared after 1884. This was when the French began a program of investment in new ships and technologies.

Finally, Proposition 5 predicts that when the fixed cost of naval investment (ϕ) is higher, there is more incentive for the naval leader to strategically overinvest in naval power. This result can help account for both Britain's major investment in its navy in the late 19th and early 20th centuries and the naval spending of the United States during the late 20th century (and especially after 1989). Both were the undisputed naval hegemony during the respective eras and yet, they continued to significantly outspend and thus increase their lead over their rivals.

Conclusion

Sea power has played a critical role in the rise and fall of world powers throughout history. Nonetheless, it has not received much attention from scholars working in historical political economy. In this paper we introduce a formal model to analyze why states invest in sea power. This model sheds light on why sea power is highly concentrated and what factors can lead to changes in the distribution of sea power.

Specifically, because investments in sea power require high fixed costs and are capital intensive, we model naval competition using a Stackelberg game. We take the existence of a dominant naval power as given and study the evolution of naval dominance. In our setup, the naval leader typically has an important advantage in its underlying naval capacity over rival powers. An increase in spending by the naval leader can thus deter other states from investing in sea power. While the model does not predict when and how naval dominance emerges, it helps to explain the persistence of a highly asymmetric distribution of sea power that we observe in modern and historical data. We also find conditions when investment by the naval follower can prompt the leader to invest more, triggering an intensification of naval competition. To test the predictions of the model, we introduce a novel dataset of ships and sea-based trade in the late 19th and early 20th centuries, and find empirical support for the model for this "age of navalism."

A Proofs for Section 3

Proofs of Propositions 1–4: When $x_1 \neq 0$, write $x_2 = nx_1$ and let player 2 choose n instead of x_2 , so

$$\pi_1(x_1) = \frac{1}{1+n}V - \theta_1 x_1^2, \quad \pi_2(n) = \frac{n}{1+n}V - \theta_2 n^2 x_1^2.$$

Player 2's best response satisfies

$$\begin{aligned}\frac{\partial \pi_2}{\partial n} &= \frac{1}{(1+n)^2} V - 2\theta_2 n x_1^2 = 0 \\ \implies \frac{1}{2\theta_2 n(1+n)^2} V &= x_1^2.\end{aligned}\quad (1)$$

The second-order condition is

$$\frac{\partial^2 \pi_2}{\partial n^2} = -\frac{2}{(1+n)^3} V - 2\theta_2 x_1^2 \text{ which is negative for } n \geq 0.$$

Hence any positive root of the FOC is optimal. Also,

$$\pi_2'(0) = V > 0, \lim_{n \rightarrow \infty} \pi_2'(n) = -\infty,$$

so there exists a unique positive n^* satisfying $\pi_2'(n^*) = 0$.

Furthermore, Equation (1) shows that $\frac{\partial n^*}{\partial x_1} < 0$ for $x_1 > 0$, so the map $x_1 \mapsto n^*$ is injective. Thus we can assume that player 1 chooses n instead of x_1 . Rewriting, player 1 maximizes

$$\pi_1(n) = \frac{1}{n+1} V - \frac{\theta_1}{2\theta_2 n(1+n)^2} V.$$

The first-order condition is

$$\begin{aligned}\frac{\partial \pi_1}{\partial n} &= -\frac{1}{(1+n)^2} V + \frac{\theta_1(3n+1)}{2\theta_2 n^2(1+n)^3} V = 0 \\ \implies -n^2 - n^3 + \frac{\theta_1}{2\theta_2}(3n+1) &= 0\end{aligned}\quad (2)$$

Let $t = \frac{\theta_1}{\theta_2}$, and let

$$f(n, t) = -n^3 - n^2 + 1.5tn + 0.5t.$$

This cubic has discriminant

$$\Delta = 13.5t^3 + 9t^2 + 2t > 0,$$

so it has roots $\alpha, \beta, \gamma \in \mathbb{R}$.

We have $\alpha + \beta + \gamma = -1 < 0$, $\alpha\beta\gamma = 0.5t > 0$, so exactly one root is positive.

$$\pi_1(0) = 0, \lim_{n \rightarrow \infty} \pi_1 = 0^+ \implies \text{the root is a maximizer.}$$

This also shows that player 1 can obtain positive payoffs for some $x_1 \neq 0$, so $x_1 = 0$ will never be chosen and the FOCs derived above always hold.

Note that

$$\frac{\partial f}{\partial t} = 3n + 1 > 0,$$

so if $n^*(t)$ is the positive root of $f(n, t)$, then for any $t_1 < t_2$,

$$0 = f(n^*(t_1), t_1) < f(n^*(t_1), t_2).$$

Since $\lim_{n \rightarrow \infty} f(n, t_2) = -\infty$, f must have a root in $(n^*(t_1), \infty)$. Thus $n^*(t_1) < n^*(t_2)$, and $\frac{dn^*}{dt} > 0$. Also, $\frac{dt}{d\theta_1} > 0$, $\frac{dt}{d\theta_2} < 0$.

By chain rule on Equation (1),

$$\frac{dx_1}{d\theta_1} = \frac{\partial x_1}{\partial n^*} \frac{dn^*}{dt} \frac{dt}{d\theta_1} + \frac{\partial x_1}{\partial \theta_2} \frac{d\theta_2}{d\theta_1} + \frac{\partial x_1}{\partial V} \frac{dV}{d\theta_1} < 0 \text{ (Proposition 1a)}.$$

Using Equations (1) and (2), we can rewrite

$$\begin{aligned} x_1^2 &= \frac{n}{\theta_1(1+n)(3n+1)}V \\ \implies \frac{\partial(x_1^2)}{\partial n} &= \frac{1-3n^2}{\theta_1(1+n)^2(3n+1)^2}V, \end{aligned} \quad (3)$$

which is positive iff $n < \frac{1}{\sqrt{3}}$, or $t < \frac{2}{3\sqrt{3}}$, by substituting $n = \frac{1}{\sqrt{3}}$ into Equation (2). So chain rule gives

$$\frac{dx_1}{d\theta_2} < 0 \iff \theta_1 < \frac{2}{3\sqrt{3}}\theta_2 \text{ (Proposition 3b)}.$$

From (2), a proportional fall in θ_1 and θ_2 that leaves t unchanged will leave $n^*(t)$ unchanged, which implies, from Equation (3), that x_1 and x_2 will increase proportionally (Proposition 4).

We next analyze $x_2^2 = n^2 x_1^2 = \frac{n^2}{2\theta_2 n(1+n)^2}V = \frac{n^3}{\theta_1(1+n)(3n+1)}V$.

The first expression gives

$$\frac{\partial(x_2^2)}{\partial n} = \frac{1-n}{2\theta_2(1+n)^3}V,$$

which is positive iff $n < 1$, or $t < 1$. So

$$\frac{dx_2}{d\theta_1} > 0 \iff \theta_1 < \theta_2 \text{ (Proposition 3a)}.$$

The second expression gives

$$\frac{\partial(x_2^2)}{\partial n} = \frac{3n^4 + 8n^3 + 3n^2}{\theta_1(1+n)^2(3n+1)^2}V,$$

which is always positive, so

$$\frac{dx_2}{d\theta_2} < 0 \text{ (Proposition 1b).}$$

From Equation (2), the equilibrium ratio n is independent of V . Thus Equation (1) gives

$$\frac{dx_1}{dV} > 0, \text{ which implies that } \frac{dx_2}{dV} > 0 \text{ (Proposition 2).}$$

□

Proof of Propositions 5: Previously we assume that investing in the military incurs no fixed cost, i.e., $\phi = 0$. Now consider the case of $\phi > 0$. The fixed cost is only incurred when a country makes a strictly positive level of military investment.

Suppose the follower will *exit* (i.e., set $x_2 = 0$ to avoid paying the fixed cost) if $x_2 > 0$ implies $\pi_2 \leq 0$.

At $\phi = 0$, both the leader and follower will make strictly positive investments. If exit is not an option for either country, as ϕ increases, it will reach a point when $\pi_2^* = 0$; let the corresponding level of the fixed cost be denoted by $\hat{\phi}$. From here on, we only consider the range $0 < \phi < \hat{\phi}$.

Consider the leader's problem at stage 1. It can choose x_1 optimally in anticipation that the follower will choose a strictly positive x_2 in stage 2, in which case x_1^* and x_2^* will be given by Equation (1) and (2) in the previous proof. Alternatively, the leader can choose to *overinvest* to the point that in stage 2, $\pi_2 \leq 0$ for any $x_2 > 0$.

If the leader does not overinvest,

$$\pi_1^* = \frac{1}{1+n^*}V - \theta_1 x_1^{*2} - \phi,$$

where x_1^* and n^* are given by Equations (1) and (2) in the previous proof.

If the leader overinvests,

$$\bar{\pi}_1 = V - \theta_1 \bar{x}_1^2 - \phi,$$

where $\bar{x}_1$ is given by $\pi_2 = \frac{\bar{n}}{1+\bar{n}}V - \theta_2 \bar{n}^2 \bar{x}_1^2 - \phi = 0$ and Equation (1) in the previous proof.

The leader overinvests if $\bar{\pi}_1 > \pi_1^*$, or

$$\left(\bar{n} - \frac{t}{\bar{n}(1+\bar{n})} - \frac{t}{2n^*(1+n^*)^2}\right)V + \frac{t}{\bar{n}^2}\phi > 0$$

It can be further shown that both $\bar{n}$ and the LHS of the inequality are increasing in ϕ . Hence, the higher the fixed cost, the less the leader needs to over invest in stage 1 to drive the follower to exit in stage 2 and the greater the incentive for the leader to over invest. $\square$

B Data Appendix

Data on European army sizes before the 20th century do not exist in a single unified dataset. We therefore compile estimates from a variety of different sources. Note that comparisons over time can be difficult as the size of armies varied between peacetime and wartime — unlike with naval strength, the size of an army could be greatly enlarged by including militia regiments or by introducing conscription. Nonetheless, we believe that the data we have compiled accurately reflects broad patterns in army sizes over time.

HHI for Armies: 1740

The estimates of army sizes at the outbreak of the conflict are originally from Kriegsarchiv and Kriegsarchiv (1896) and available on Wikipedia:

https://en.wikipedia.org/wiki/War_of_the_Austrian_Succession.

HHI for Armies: 1800

Sources: for the Russia, Prussia, Great Britain, France, and Austria we rely on Rothenberg (1978). For the Ottoman empire we rely on Johnson and Bell (1988).

Among the secondary powers. The army of Savoy was one of the largest and most professional in Europe. Its peacetime strength in 1786 was 38,997. However, in wartime, it mobilized 110,000 men in the 1790s (*The Sardinian Army: A Survey* n.d.). For the size of the Swedish army in 1805 we use Olofsson (2008); for the size of the Dutch army we use the size of the Batavian Army in 1799 which was 25,000 according to Street and Overmeire (2014). Our estimate for the size of the Danish army (which may be on the low side) is based on the discussion of Wolter (n.d.).

HHI for Armies: 1914

For a consistent set of estimates for army sizes on the eve of World War I we used the *Encyclopedia Britannica*: <https://www.britannica.com/event/World-War-I/Forces-and-resources-of-the-combatant-nations-in-1914>.

HHI for Navies

Sources: For 1500–1860 we construct HHI using global power warships measured by Modelski and Thompson (1988). For 1861–1914 we construct HHI using total displacement reported in Brassey's Annuals.

Numbers of ships and displacement data

Source: Brassey, Thomas Brassey. 1886–1913. *Brassey's annual*. New York: Praeger Publishers.

GDP and population data

Source: The Maddison-Project, <http://www.ggdc.net/maddison/maddison-project/home.htm>, 2013 version.

Military capacity index

Source: Singer, J. David, Stuart Bremer, and John Stuckey (1972). "Capability Distribution, Uncertainty, and Major Power War, 1820–1965" in Bruce Russett (ed) *Peace, War, and Numbers*, Beverly Hills: Sage, 19–48.

Trade data

Sources: 1) J. Ernesto López-Córdova and Christopher M. Meissner. 2008. "The Impact of International Trade on Democracy: A Long-Run Perspective" *World Politics* Vol. 60, No. 4, pp. 539–575;

2) Katherine Barbieri, "Economic Interdependence and Militarized Interstate Conflict, 1870–1985" (Ph.D. diss., Binghamton University).

References

Allen, D. W. 2002. "The British Navy Rules: Monitoring and Incompatible Incentives in the Age of Fighting Sail". *Explorations in Economic History* 39(2): 204–31.

- Benjamin, D. K. and C. Thornberg. 2007. "Organization and Incentives in the Age of Sail". *Explorations in Economic History* 44(2): 317–41.
- Brewer, J. 1988. *The Sinews of Power*. Cambridge, MA: Harvard University Press.
- Corbett, J. S. 1911. *Some Principles of Maritime Strategy*. London: The Project Gutenberg eBook.
- Dixit, A. K. and R. S. Pindyck. 1994. *Investment under Uncertainty*. Princeton, NJ: Princeton University Press.
- Dull, J. R. 2009. *The Age of the Ship of the Line: The British & French Navies, 1650–1815*. Lincoln, Nebraska: University of Nebraska Press.
- Falk, K. L. 2000. *Why Nations Put to Sea: Technology and the Changing Character of Power in the Twenty-First Century*. New York: Garland Publishing.
- Findlay, R. and K. H. O'Rourke. 2007. *Power and Plenty*. Princeton, New Jersey: Princeton University Press.
- Garfinkel, M. R. and S. Skaperdas. 2007. "Economics of Conflict: An Overview". In: *Handbook of Defense Economics*. Ed. K. Hartley and T. Sandler. Vol. 2. *Handbook of Defense Economics*. Elsevier. Chap. 22, 649–709.
- Glaser, D. J. and A. S. Rahman. 2016. "Ex Tridenti Mercatus? Sea-Power and Maritime Trade in the Age of Globalization". *Journal of International Economics* 100(C): 95–111.
- Glete, J. 2000. *Warfare at Sea, 1500–1650*. London: Routledge.
- Gray, C. S. 1992. *The Leverage of Sea Power*. New York: The Free Press.
- Harding, R. 1999. *Seapower and Naval Warfare, 1650–1830*. London: UCL Press.
- Helpman, E. 1988. "Growth, Technological Progress, and Trade". *Empirica* 15(1): 5–25.
- Herman, A. 2004. *To Rule the Waves*. New York: Harper Collins Publishers.
- Hirshleifer, J. 2001. *The Dark Side of the Force: Economic Foundations of Conflict Theory*. Cambridge: Cambridge University Press.
- Iambi, I. N. 1984. *The Navy and German Power Politics, 1862–1914*. New York: Routledge.
- Johnson, N. D. and M. Koyama. 2017. "States and Economic Growth: Capacity and Constraints". *Explorations in Economic History* 64: 1–20.
- Johnson, W. E. and C. Bell. 1988. *The Ottoman Empire and the Napolonic Wars*. Leeds: Raider Books.
- Kennedy, P. 1976. *The Rise and Fall of British Naval Mastery*. London: Penguin.
- Kriegsarchiv, K. and K. Kriegsarchiv. 1896. *Oesterreichischer Erbfolgekrieg*. Vol. I. Vienna: Verlag L. W. Seidel & Sohn.
- Lambert, A. 2018. *Seapower States*. New Haven: Yale University Press.
- Lewis, A. R. 1951. *Naval Power and Trade in the Mediterranean*. Princeton, NJ: Princeton University Press.

- Lindsey, D. 2019. "Mutual Optimism and Costly Conflict: The Case of Naval Battles in the Age of Sail". *Journal of Politics* 81(4): 1181–96.
- López-Córdova, J. E. and C. M. Meissner. 2008. "The Impact of International Trade on Democracy: A Long-Run Perspective". *World Politics* 60(4): 539–75.
- Mackinder, H. J. 1904. "The Geographical Pivot of History". *The Geographical Journal* 23: 421–37.
- Maddison, A. 2003. *The World Economy: Historical Statistics*. Paris, France: OECD.
- Mahan, T. 1890. *The Influence of Sea Power Upon History: 1660–1783*. Boston: Little Brown and Company.
- Marcus, G. J. 1971. *The Age of Nelson, The Royal Navy 1793–1815*. New York: The Viking Press.
- Marder, A. J. 1961. *From the Dreadnought to Scapa Flow: The Royal Navy in the Fisher Era, 1904–1919*. New York: Oxford University Press.
- McEvedy, C. and R. Jones. 1978. *Atlas of World Population History*. Harmondsworth: Penguin.
- Modelski, G. and W. R. Thompson. 1988. *Seapower in Global Politics, 1494–1993*. London: Macmillan Press.
- O'Hara, G. 2010. *Britain and the Sea Since 1600*. Basingstoke: Palgrave MacMillan.
- Olofsson, M. 2008. "The Swedish Army in the Napoleonic Wars". https://www.napoleon-series.org/military-info/organization/Sweden/Organization/c_swedisharmy1.html. Accessed: 2020-12-14.
- Padfield, P. 1974. *The Great Naval Race: Anglo-German Naval Rivalry 1900–1914*. New York: David McKay Company.
- Padfield, P. 1982. *Tides of Empires: Decisive Naval Campaigns in the Rise of the West, Volume 2*. London: Routledge & Kegan Paul.
- Parker, G. 1976. "The "Military Revolution," 1560–1660 — A Myth?" *Journal of Modern History* 48(2): 195–214.
- Parker, G. 1988. *The Military Revolution: Military Innovation and the Rise of the West, 1500–1800*. Cambridge: Cambridge University Press.
- Rahman, A. S. 2010. "Fighting the Forces of Gravity: Seapower and Maritime Trade between the 18th and 20th Centuries". *Explorations in Economic History* 47(1): 28–48.
- Richmond, S. H. 1934. *Sea Power in the Modern World*. London: G. Bell & Sons LTD.
- Rodrik, D. 1995. "Getting Interventions Right: How South Korea and Taiwan Grew Rich". *Economic Policy* 10(20): 53–107.
- Romer, D. H. and J. A. Frankel. 1999. "Does Trade Cause Growth?" *American Economic Review* 89(3): 379–99.
- Rose, L. A. 2007. *Power at Sea: The Age of Navalism 1890–1918*. Vol. 1. Columbia, Missouri: University of Missouri Press.

- Rothenberg, G. E. 1978. *The Art of Warfare in the Age of Napoleon*. Bloomington: Indiana University Press.
- Seligmann, M. S. 2015. *The Royal Navy and the German Threat 1901–1914*. Oxford: Oxford University Press.
- Seligmann, M. S. 2016. “The Anglo-German Naval Race, 1898–1914”. In: *Arms Races in International Politics: From the Nineteenth to the Twenty-First Century*. Ed. T. Mahnken, J. Maiolo, and D. Stevenson. Oxford: Oxford University Press, 21–40.
- Shepard, A. M. 1925. *Sea Power in Ancient History*. London: William Heinemann Ltd.
- Skaperdas, S. 1996. “Contest Success Functions”. *Economic Theory* 7(2): 283–90.
- Sprout, H. and M. Sprout. 1939. *The Rise of American Naval Power, 1776–1918*. Princeton, NJ: Princeton University Press.
- Starr, C. G. 1989. *The Influence of Sea Power on Ancient History*. Oxford: Oxford University Press.
- Strading, R. 1979. “Catastrophe and Recovery: The Defeat of Spain, 1639–43”. *History* 64(211): 205–19.
- Street, G. and K. V. Overmeire. 2014. *Dutch Troops of the French Revolutionary and Napoleonic Wars 1793–1810*. Eastwood: Partizan Press.
- The Sardinian Army: A Survey*. n.d. <https://napoleoninpiedmont.weebly.com/the-sardinian-army-a-survey.html>.
- Till, G. 2013. *Seapower: A Guide for the Twenty-First Century*. London: Routledge.
- Voth, H.-J. and G. Xu. 2020. “Patronage for Productivity: Selection and Performance in the Age of Sail”. CEPR Discussion Papers 13963.
- Wolter, H. C. n.d. *Danish Infantry of the Line and Light Infantry 1803–1814: The Perry Achievement*. <https://www.chakoten.dk/wp-content/uploads/Danish-Infantry-of-the-Line-and-Light-Infantry-1803-1814.-The-Perry-Achievement-.pdf>.